

National University of Computer and Emerging Sciences

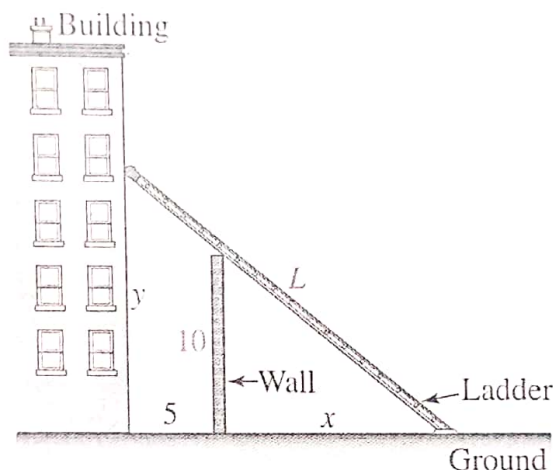
FAST School of Computing

Fall-2021

Islamabad Campus

Question 1 [10Marks]

A 10-ft wall stands 5ft from a building. A ladder, supported by the wall, touches the building as shown in the figure. Express the length of the ladder as a function of x shown in the figure. Also find the domain of the function.



Let L denotes the length of ladder. With ' x ' & ' y ' defined in the above given figure we see that there are two right triangles; the larger triangle has three sides with length L , y & $x+5$ & the smaller triangle has two sides of length x & 10 . The ladder is the hypotenuse of larger right triangle, so by Pythagorean theorem

$$L^2 = (x+5)^2 + y^2 \rightarrow (i)$$

The right triangles in the figure are similar because they both contain a right angle & share the common acute angle the ladder makes with the ground. We use the fact that the ratios of corresponding sides of similar triangles are equal.

$$\frac{y}{x+5} = \frac{10}{x} \rightarrow (ii)$$

Solving (ii) for y in terms of x gives $y = \frac{10(x+5)}{x}$ so (i) becomes

$$L^2 = (x+5)^2 + \left(\frac{10(x+5)}{x}\right)^2$$

$$L^2 = (x+5)^2 \left[1 + \frac{100}{x^2}\right]$$

$$L^2 = (x+5)^2 \left[\frac{x^2 + 100}{x^2}\right]$$

$$L(x) = \frac{x+5}{x} \sqrt{x^2 + 100}$$

Domain of $L(x)$ is $(0, \infty)$

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Question 2 [10Marks]

The radius of a right circular cylinder is increasing at the rate of 4 cm/sec but its total surface area remains constant at $600\pi \text{ cm}^2$. At what rate is the height changing when the radius is 10cm ?

$$\text{Radius} = r = 10\text{cm}, \frac{dr}{dt} = 4 \text{ cm/s}, \frac{dh}{dt} = ?, \frac{dS}{dt} = 0$$

$$\text{Surface Area} = S = 2\pi r^2 + 2\pi rh \rightarrow \textcircled{i}$$

(2 circular ends) (cylindrical wall)

Differentiating $\textcircled{i}$ w.r.t time we get

$$\frac{dS}{dt} = 4\pi r \frac{dr}{dt} + 2\pi r \frac{dh}{dt} + 2\pi h \frac{dr}{dt} \rightarrow \textcircled{ii}$$

$$S = 600\pi = 2\pi(10)^2 + 2\pi(10)h$$

$$600\pi = 200\pi + 20\pi h$$

$$400\pi = 20\pi h$$

$$\boxed{h = 20}$$

Substituting values in $\textcircled{ii}$ we get

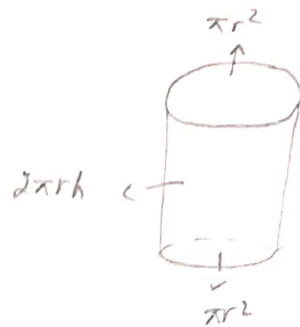
$$0 = 4\pi(10)(4) + 2\pi(10) \frac{dh}{dt} + 2\pi(20)(4)$$

$$0 = 160\pi + 20\pi \frac{dh}{dt} + 160\pi$$

$$0 = 320\pi + 20\pi \frac{dh}{dt}$$

$$\frac{dh}{dt} = -16 \text{ cm/s}$$

So the height is changing at a rate of -16 cm/s .



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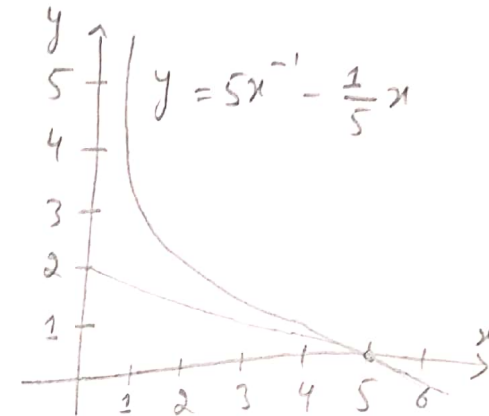
Question 3 [10Marks]

Find the area of the triangle formed from the coordinate axes and the tangent line to the curve $y = 5x^{-1} - \frac{1}{5}x$ at the point $(5, 0)$.

The derivative of y w.r.t x is

$$y'(x) = \frac{d}{dx} [5x^{-1}] - \frac{d}{dx} [\frac{1}{5}x]$$

$$y'(x) = -5x^{-2} - \frac{1}{5}$$



The slope of tangent line at the point $(5, 0)$ is $y'(5) = -\frac{2}{5}$
The eq of tangent line at this point is

$$y - 0 = -\frac{2}{5}(x - 5)$$

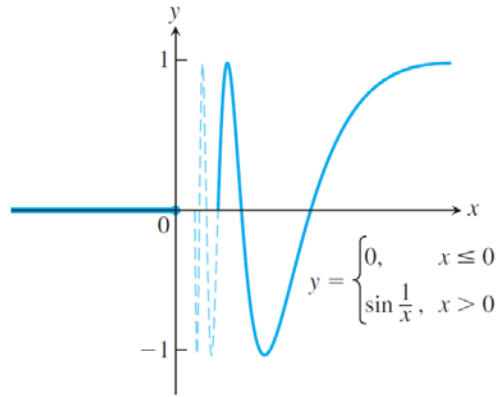
$$y = -\frac{2}{5}x + 2$$

Since the y-intercept of this line is 2, the right triangle formed from the coordinate axes & tangent line has legs of length 5 & 2, so its area is $\frac{1}{2}(5)(2) = 5$.

Question 4A

Let

$$f(x) = \begin{cases} 0 & , \quad x \leq 0 \\ \sin \frac{1}{x} & , \quad x > 0 . \end{cases}$$



a. Does $\lim_{x \rightarrow 0^+} f(x)$ exist? If so, what is it? If not, why not?

b. Does $\lim_{x \rightarrow 0^-} f(x)$ exist? If so, what is it? If not, why not?

c. Does $\lim_{x \rightarrow 0} f(x)$ exist? If so, what is it? If not, why not?

Solution:-

a. No. The limit $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sin \frac{1}{x}$ does not exist since $\sin \frac{1}{x}$ does not approach any single value as x approaches 0.

b. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (0) = 0$.

c. No. The limit $\lim_{x \rightarrow 0} f(x)$ does not exist as $\lim_{x \rightarrow 0^+} f(x)$ does not exist.

Question 4B

Show that

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & , \quad x \neq 0 \\ 0 & , \quad x = 0 \end{cases}$$

is differentiable at $x = 0$ and find $f'(0)$.

Solution: One has to show that the given function $f(x)$ is differentiable at $x = 0$. By definition

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

Here $x_0 = 0$, so

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \left[\frac{1}{h} \cdot \left\{ h^2 \sin\left(\frac{1}{h}\right) - 0 \right\} \right] = \lim_{h \rightarrow 0} \frac{h^2}{h} \sin\left(\frac{1}{h}\right).$$

$$f'(0) = \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right). \quad (1)$$

As $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$, it follows that

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|.$$

Since

$$\lim_{x \rightarrow 0} -|x| = 0 \text{ and } \lim_{x \rightarrow 0} |x| = 0,$$

the Sandwich theorem implies that

$$\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0. \quad (2)$$

From equation (2), equation (1) implies that

$$f'(0) = \lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right) = 0.$$

Hence, the given function is differentiable at $x = 0$ with $f'(0) = 0$. Alternatively, one can use the concept of one-sided derivatives as well.

Q 5A:

$$y = \sin u^2$$

$$u = (p^2 + 1)^3$$

$$p = \frac{1}{x^2}$$

$$\Rightarrow u = \left(\left(\frac{1}{x^2} \right)^2 + 1 \right)^3 = \left(\frac{1}{x^4} + 1 \right)^3$$

$$\Rightarrow \frac{1}{x^4} = \frac{1}{x^4}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dp} \cdot \frac{dp}{dx} \quad (\text{Chain rule})$$

$$\frac{dy}{dx} = \frac{d}{du} (\sin u^2) \cdot \frac{d}{dp} (p^2 + 1)^3 \cdot \frac{d}{dx} \left(\frac{1}{x^2} \right)$$

$$\frac{dy}{dx} = \cos u^2 (2u) \left[3(p^2 + 1)^2 (2p) \right] \left[\frac{-2}{x^3} \right]$$

$$\frac{dy}{dx} = 2u \cos u^2 (6p (p^2 + 1)^2) \left(\frac{-2}{x^3} \right)$$

$$\frac{dy}{dx} = \frac{-24 u \cos u^2 [p (p^2 + 1)^2]}{x^3}$$

$$\frac{dy}{dx} = \frac{-24}{x^3} \left(\frac{1}{x^4} + 1 \right)^3 \cos \left(\frac{1}{x^4} + 1 \right)^6 \cdot \frac{1}{x^2} \left(\frac{1}{x^4} + 1 \right)^2$$

$$\boxed{\frac{dy}{dx} = \frac{-24}{x^5} \left(\frac{1}{x^4} + 1 \right)^5 \cos \left(\frac{1}{x^4} + 1 \right)^6}$$

Ans

Q5B:

$$S = 4.5t^2$$

(i) Average velocity over the interval $[0, 12]$

$$V_{av} = \frac{S(12) - S(0)}{12 - 0}$$

$$= \frac{4.5(12)^2 - 4.5(0)^2}{12}$$

$$= \frac{648}{12}$$

$$\boxed{V_{av} = 54 \text{ ft/sec}} \quad \underline{\underline{\text{Ans}}}$$

(ii) Instantaneous velocity of the car at $T=6$

$$\left. \frac{ds}{dt} \right|_{T=6} = 4.5(2t)$$
$$= 9t$$

at $T=6$

$$= 9(6)$$

$$\boxed{\text{Instantaneous velocity} = 54 \text{ ft/sec}} \quad \underline{\underline{\text{Ans}}}$$

Q6(a). Find the limit (if exists) of following rational function by using limit laws. If limit does not exist, give reason.

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{|x| - 3}$$

Soln

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{|x| - 3}$$

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x^2 - 9}{|x| - 3}$$

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \begin{cases} \frac{x^2 - 9}{x - 3} & x \geq 3 \\ \frac{x^2 - 9}{-x - 3} & x < 3 \end{cases}$$

$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$

Simplify!

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \begin{cases} \frac{(x-3)(x+3)}{x-3} & x \geq 3 \\ \frac{(x-3)(x+3)}{-(x+3)} & x < 3 \end{cases}$$

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \begin{cases} (x+3) & x \geq 3 \\ -(x-3) & x < 3 \end{cases}$$

Applying limit

From left

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} -(x-3)$$

$$= - \lim_{x \rightarrow 3^-} x + \lim_{x \rightarrow 3^-} 3$$

$$= -3 + 3$$

$$\lim_{x \rightarrow 3^-} f(x) = 0$$

Left Hand Limit
→ (1)

From Right.

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (x+3)$$

$$= \lim_{x \rightarrow 3^+} x + \lim_{x \rightarrow 3^+} (3)$$

$$= 3 + 3$$

$$\lim_{x \rightarrow 3^+} f(x) = 6$$

Right Hand Limit
→ (2)

Limit Does not exist, Because the left hand limit (Eq 1) & Right hand limit (Eq 2) are not equal

(2)

b) Use implicit differentiation to find slope of tangent line to the curve at specified points (3, 2).

$$2^x + y^2 = 2xy$$

Soln $2^x + y^2 = 2xy$
slope of tangent line $= \frac{dy}{dx} = ?$

$$\frac{d}{dx}(2^x + y^2) = \frac{d}{dx}(2xy) \text{ product rule.}$$
$$2^x \ln(2) + 2y \frac{dy}{dx} = 2 \left(x \cdot \frac{dy}{dx} + 1 \cdot y \right)$$
$$2^x \ln(2) + 2y \frac{dy}{dx} = 2x \frac{dy}{dx} + 2y$$
$$2y \frac{dy}{dx} - 2x \frac{dy}{dx} = 2y - 2^x \ln(2)$$
$$\frac{dy}{dx} = \frac{2y - 2^x \ln(2)}{2y - 2x}$$

At (3, 2)

$$\left. \frac{dy}{dx} \right|_{(3,2)} = \frac{2(2) - 2^{(3)} \ln(2)}{2(2) - 2(3)}$$
$$= \frac{4 - 8 \ln 2}{4 - 6}$$
$$\left. \frac{dy}{dx} \right|_{(3,2)} = \frac{4 - 8 \ln 2}{-2}$$

Q # 7

$$\text{Revenue} = r(x) = x p = x \left(3 - \frac{x}{40} \right)^2$$

02

$$0 \leq x \leq 60$$

$$\frac{dr}{dx} = \left(3 - \frac{x}{40} \right)^2 + x \cdot 2 \left(3 - \frac{x}{40} \right) \left(-\frac{1}{40} \right)$$

$$\frac{dr}{dx} = \left(3 - \frac{x}{40} \right) \left[3 - \frac{x}{40} - \frac{2x}{40} \right]$$

$$\frac{dr}{dx} = \left(3 - \frac{x}{40} \right) \left[\frac{120 - 3x}{40} \right]$$

03

$$\frac{dr}{dx} = 0 \Rightarrow x = 40, x = 120$$

01

Since $x = 120$ does not belong to the domain
when 40 people are in the bus the marginal
revenue is zero and $P(40) = \left(3 - \frac{40}{40} \right)^2 = 4$ \$.

part (b) $f(x) = \frac{8}{x^2 - 4}$

01

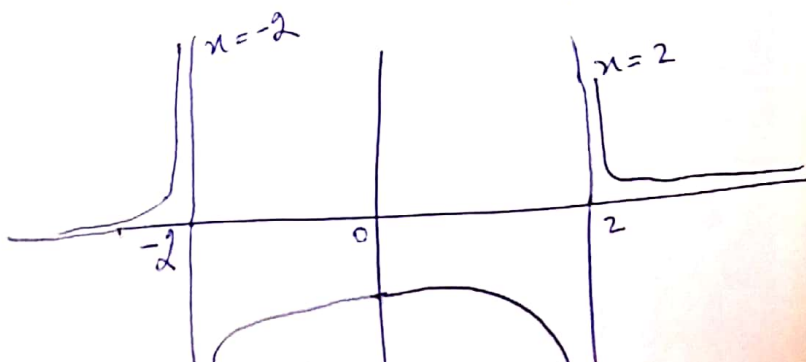
vertical asymptote $\Rightarrow x = 2, -2$

horizontal

$$\lim_{x \rightarrow \infty} \frac{8}{x^2 - 4} = 0$$

$$\Rightarrow y = 0$$

01



02

Q # 8

$$(x-h)^2 + (y-k)^2 = a^2 \rightarrow (1)$$

$$y = x^2 + 1, \quad y' = 2x, \quad y'' = 2 \quad \underline{01}$$

Differentiating eq (1)

$$2(x-h) + 2(y-k)y' = 0.$$

Again differentiating

$$2 + 2[y'^2 + (y-k)y''] = 0.$$

$$2(y-k)y'' = -2 - 2y'^2$$

$$y'' = -(1 + y'^2)/(y-k) \quad \underline{03}$$

at the point (1, 2), $y' = y'' = 2$

$$2 = \frac{-(1+4)}{(2-k)} \Rightarrow 4 - 2k = -5$$

$$\Rightarrow -2k = -9$$

$$\boxed{k = 9/2} \quad \underline{02}$$

Also both curves are tangent at (1, 2)

$$\text{so slope} = 2x|_{(1,2)} = 2$$

tangent line $\Rightarrow \boxed{y = 2x}$ line containing

(h, k) and (1, 2) is perpendicular to $y = 2x$

$$\frac{k-2}{h-1} = -\frac{1}{2} \Rightarrow 2k - 4 = -h + 1$$

$$\Rightarrow \boxed{h = 5 - 2k = 5 - 2(9/2) = -4} \quad \underline{03}$$

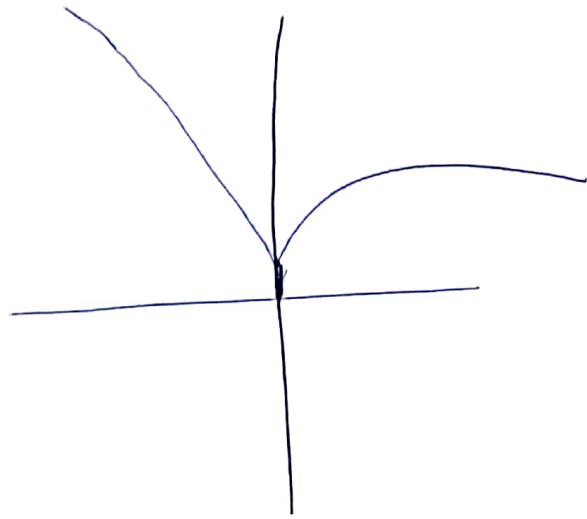
Since (1, 2) lies on circle so $a = \frac{5\sqrt{5}}{2} \quad \underline{01}$

Q # 9

$$y = 4x^{2/5} - 2x$$

03

- (a) 02 graph appears to have a cusp at $x=0$.



- (b) $f(0) = 0$
 $f(0+h) = 4h^{2/5} - 2h$

05

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \frac{4h^{2/5} - 2h - 0}{h}$$

$$= \lim_{h \rightarrow 0} (4h^{2/5-1} - 2) = \lim_{h \rightarrow 0} \left(\frac{4}{h^{3/5}} - 2 \right) = \infty$$

$$\lim_{h \rightarrow 0^-} \left(\frac{4}{h^{3/5}} - 2 \right) = -\infty$$

no vertical tangent at $x=0$.